

NETWORK SECURITY

Lab Project Summary Report

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Objective

This lab project provided hands-on experience in network reconnaissance, attack simulation, firewall configuration, network address translation (NAT), and secure network design. Conducted across six progressive assignments, the group built practical skills in defending Linux-based servers against real-world attack scenarios while implementing layered security controls.

Learning Goals

- Discover and enumerate live hosts and open services on a local network using active scanning.
- Deploy and verify HTTP and FTP services on a Linux server, then confirm port assignments using multiple tools.
- Understand attack traffic patterns (port scans, ICMP floods, Smurf attacks) by observing them live with traffic analysis tools.
- Design and implement iptables firewall rules to control access by IP, MAC address, protocol, and connection count.
- Apply DNAT and SNAT rules to enable port forwarding and IP masquerading through a Linux firewall.
- Design a segmented hospital network with dedicated subnets, VLANs, and layered firewall zones.

Tools Used

Tool	Purpose and Usage in thus Project
Nmap	Host discovery (ARP ping scan, -PR flag), port scanning with service detection (-sV, -sS, -A, -Pn). Used to identify open ports 21 (FTP), 22 (SSH), 80 (HTTP) and to verify firewall rules after each change.
Iptables	Core firewall tool. Used to enforce access control by source IP, MAC address, protocol, and connection limit. Implemented INPUT/OUTPUT/FORWARD rules, DNAT for port forwarding, and SNAT/MASQUERADE for IP hiding.

Tcpdump	Packet-level capture during attack scenarios. Confirmed ICMP flood traffic and baseline-captured traffic to .pcap files for comparison analysis.
Iptraf-ng	Real-time traffic monitoring. Provided visual evidence of scan patterns (RST flood = port scan) and traffic volume differences before vs. during ICMP flood attacks.
Hping3	Attack simulation tool. Generated ICMP ping floods with custom packet sizes (1000 bytes) and Smurf attacks targeting broadcast addresses to stress-test the victim machine.
Netstat	Listed active listening ports (-tulpn) to confirm services were running and bound to expected addresses after enabling/disabling FTP and HTTP.
Nload	Monitored live network bandwidth (Incoming/Outgoing bit/s). Clearly showed traffic spike from near 0 to ~1.49 MBit/s during ICMP flood attacks.
FileZilla	FTP client used on Windows to connect to the Linux FTP server. Verified iptables rules by testing connections from allowed vs. blocked hosts.
Vsftpd/Apache2	Server software installed on Linux. Apache2 served HTTP on port 80; vsftpd served FTP on port 21. Both were started, stopped, and re-verified via nmap.
Cisco Pacjet Tracer	Network simulation tool used for the hospital network design assignment. Configured VLANs, routers with sub-interfaces, ACLs, and NAT.

Key Outcome

Outcome	Status
Live host discovered on subnet 192.168.43.0/24 using nmap ARP scan (-PR flag) without prior knowledge of the target IP.	Achieved
HTTP (port 80) and FTP (port 21) services successfully installed, started, and confirmed via nmap and netstat.	Achieved
FTP service successfully stopped; nmap confirmed port 21 disappeared from scan results while port 80 remained open.	Achieved
ICMP flood attack (hping3 -1 --flood, 1000-byte packets) demonstrated via nload traffic spike from ~0 to 1.49 MBit/s on victim interface.	Achieved
Smurf attack simulated using broadcast address; traffic captured in iptraf and tcpdump before and after iptables mitigation.	Achieved
iptables rule successfully dropped ICMP flood traffic from attacker IP; traffic counters dropped to near zero after rule application.	Achieved
Selective access control: friend's Windows host restricted to HTTP only (FTP blocked); own Windows granted both HTTP and FTP access.	Achieved
ICMP reply policy: own Windows received ping replies; friend's Windows blocked; friend's Linux allowed ping but blocked from ping flooding.	Achieved

SSH connection limit: friend's Windows capped at 3 concurrent SSH sessions (--connlimit-above 3 rule); own Windows unrestricted.	Achieved
SSH MAC-address whitelisting: Linux server accepted SSH only from the specific Windows MAC (14:AC:60:C9:D9:75); all other sources timed out.	Achieved
DNAT port forwarding configured: external traffic on port 80 and port 10000 redirected to internal Windows web server via Linux firewall IP.	Achieved
SNAT/masquerade configured: friend's browser saw firewall IP (10.188.199.185) instead of real client IP when accessing web server.	Achieved
Hospital network designed in Packet Tracer with 3 user subnets (Finance/Medical, Nursing/Lab/Paramedical, Guest), server DMZ, VLANs, and ACLs.	Achieved
Asset priority classification completed: High (Finance DB, Medical DB), Medium (Lab, Nursing, Paramedical), Low (Web, Email, Guest Portal).	Achieved

Lesson Learned

The most important insight from this project is that network security is not a single tool or rule it is a layered, cumulative discipline. Each assignment built on the previous one, illustrating how an attacker's reconnaissance (nmap scan) leads directly to exploitation opportunities, and how defenders must counter at every layer.

Key Technical Insights	Security Design Principles Applied
Nmap -Pn is critical when ICMP is blocked-many scans fail silently without it.	Principle of Least Privilege: every host was given only the minimum access needed.
iptables rules are processed top-down; rule order is decisive (e.g. DROP before ACCEPT).	Defense in Depth: firewall rules, MAC filtering, and connection limits combined.
RST-flagged connections in iptraf are a reliable indicator of active port scanning.	Network Segmentation: high-value assets (Finance, Medical) isolated to dedicated subnets.
DNAT and SNAT work together to completely hide internal topology from external clients.	Verify, don't assume: every rule was confirmed with nmap and traffic tools after application.